EXP – 3 DIFFERENCE BETWEEN CLI,GUI AND VUI

CLI – COMMAND LINE INTERFACE

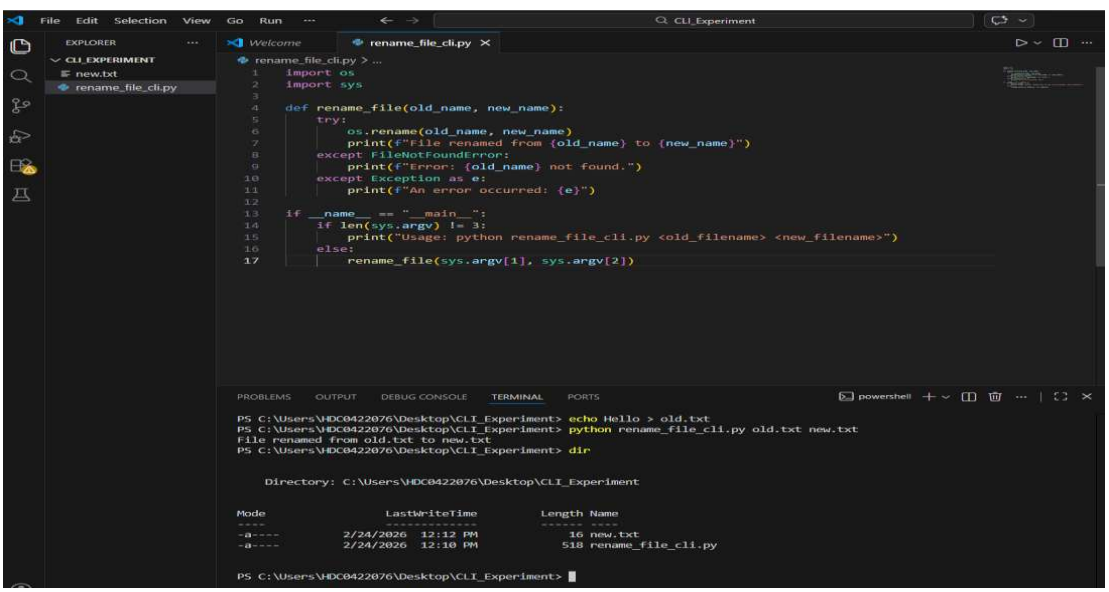
Aim:

To perform file rename operation using Command Line Interface in VS Code terminal.

Procedure:

- Open VS Code.
- Open the project folder.
- Click **Terminal → New Terminal**.
- Create a file using: echo Hello > test.txt
- Rename the file using command ren old.txt new.txt (Windows) or mv old.txt new.txt (Linux)
- Verify file name change using dir or ls.

Implementation:



Observation:

The rename command was executed successfully in the terminal. The file name changed from *old.txt* to *new.txt*. If the file name was typed incorrectly, the system displayed an error message.

Result:

File rename operation was successfully performed using CLI in VS Code.

GUI – GRAPHICAL USER INTERFACE

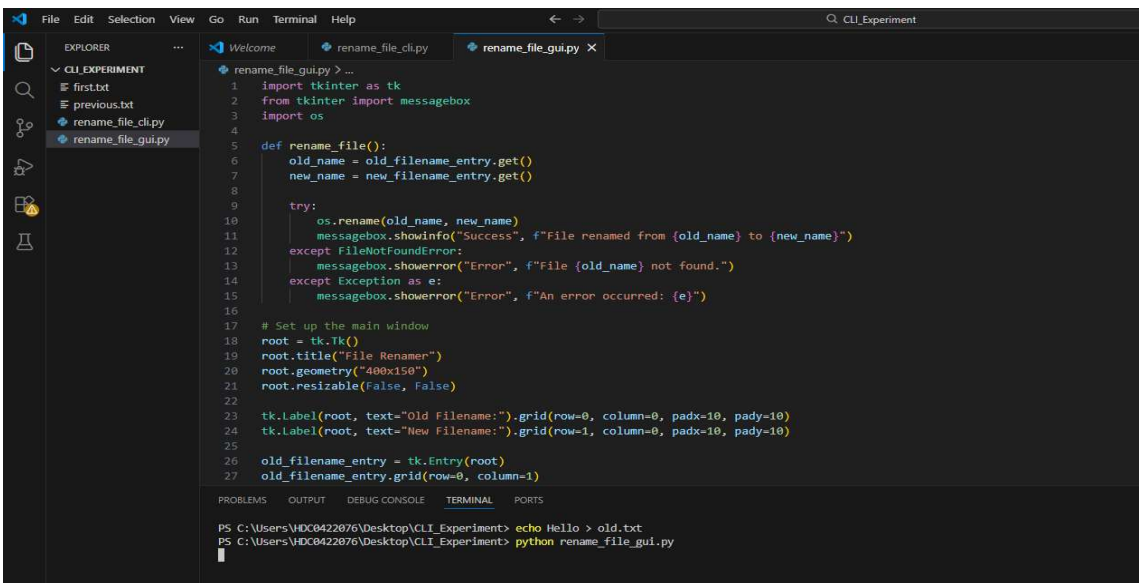
Aim:

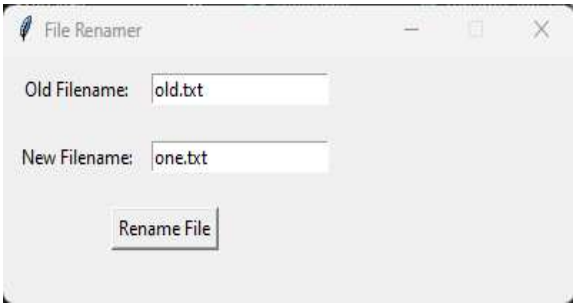
To perform file rename operation using graphical method in VS Code environment.

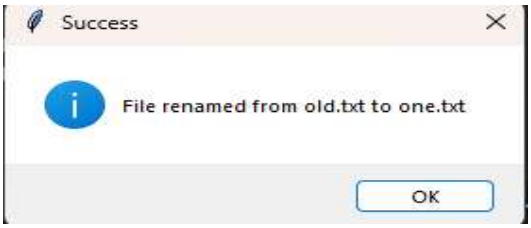
Procedure:

1. Open VS Code.
2. In the Explorer panel, create a file named *old.txt*.
3. Right-click the file.
4. Select **Rename** option.
5. Change the name to *one.txt*.
6. Press Enter to confirm.

Implementation:







Observation:

The file name was changed visually in the Explorer panel. The system provided immediate graphical feedback. No command typing was required.

Result:

File rename operation was successfully performed using GUI in VS Code.

VUI – VOICE USER INTERFACE

Aim:

To perform file rename operation using voice command in VS Code using Python.

Procedure:

1. Install required libraries: SpeechRecognition and PyAudio.
2. Write Python program to capture microphone input.
3. Execute the program in VS Code terminal.
4. Speak the command clearly:
   "rename test.txt to first dot text"
5. Observe the file name change in project folder.

Implementation:

```
import speech_recognition as sr

import os

r = sr.Recognizer()

with sr.Microphone() as source:

  print("Say a command...")

  r.adjust_for_ambient_noise(source)

  audio = r.listen(source)
```

```
try:
    command = r.recognize_google(audio).lower()
 print("You said:", command)
    if "open Notepad" in command:
        os.system("Notepad")
    elif "rename" in command and "to" in command:
        parts = command.split()
        old_name = parts[1]
        new_name = parts[3]
        os.rename(old_name, new_name)
        print("File renamed successfully")
    else:
        print("Command not recognized")
except Exception as e:
    print("Error:", e)
```

output:

```
[Running] python -u "c:\Users\bakki\OneDrive\Desktop\UID\vui_project.py"
Listening for command to rename a file...
Command received: rename test.txt to first dot text
File renamed from test.txt to first dot text

[Done] exited with code=0 in 9.789 seconds
```

Observation:

The system captured the voice input and converted it into text. When the command format was correct, the file was renamed successfully. If the command was unclear or incorrect, the system displayed an error message.

Result:

File rename operation was successfully performed using Voice User Interface in VS Code.